Frequency Analysis of UAV Vibration Anomalies Due to Wind (INTERN NOTES)

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1 Introduction

Unmanned Aerial Vehicles (UAVs) are becoming increasingly common in aerospace applications, including aerial photography, disaster relief, and the military. While the potential for smaller aircraft offers numerous benefits, unique challenges often arise with their safety precisely due to their size. For example, larger manned aircraft typically incorporate numerous redundancies in the monitoring and control systems to ensure safety, but the size of many UAVs prevents this [11]. Therefore, it is critical to develop effective health monitoring systems for UAVs, as backup systems are often not an option for these aircraft.

Fault detection and diagnosis is a valuable approach to health monitoring for these aircraft, and can be applied to a variety of components to assess the state of the aircraft and determine a need for maintenance or even flight termination. Over the last several years, fault detection and diagnosis through the analysis of vibration signals from UAVs has become an increasingly popular area of study [2, 6]. Applications of such analysis can range from diagnosing or predicting specific failure modes, such as bearing faults [5], propeller damage and friction [2, 7], excessive payload [6], to simply determining the optimal location for instrument placement to minimize the effects of nominal vibrations [20].

Frequency analysis is a common approach to understanding vibration signals originating from rotating parts, such as the motors and propellers in aircraft, especially for fault analysis, as it can reveal features in the frequency domain that arise as a result of changes in the usual periodic behavior of these components [5]. There are a wide variety of approaches to frequency analysis, including the classical Fourier transform as well as the newer wavelet decompositions and variations on each of these in an effort to resolve issues with loss of time information and cross-term interference [10, 16, 17].

In the context of System Wide Safety, the goal is to develop such systems to aid in monitoring the health of UAVs. When looking at a vibration signal to extract information on vehicle health, there is a three stage classification system, shown in Figure 1. Upon measuring the vibration signal, the first stage of classification is to determine if the vibration is a nominal vibration resulting from expected sources in the aircraft, such as motors and

propellers, or if it is an anomaly. If the vibration is classified as an anomaly, the second stage is to determine if the cause is internal or external to the aircraft. When the anomaly is due to a fault with the aircraft, the third and final stage is to diagnose what caused the anomaly, such as a fault with the motor, propeller, or sensors. However, before attempting to diagnose a fault within the aircraft, causes outside of the system health should be eliminated. Therefore, of particular interest in this study is determining when anomalies are due to external factors, specifically wind.

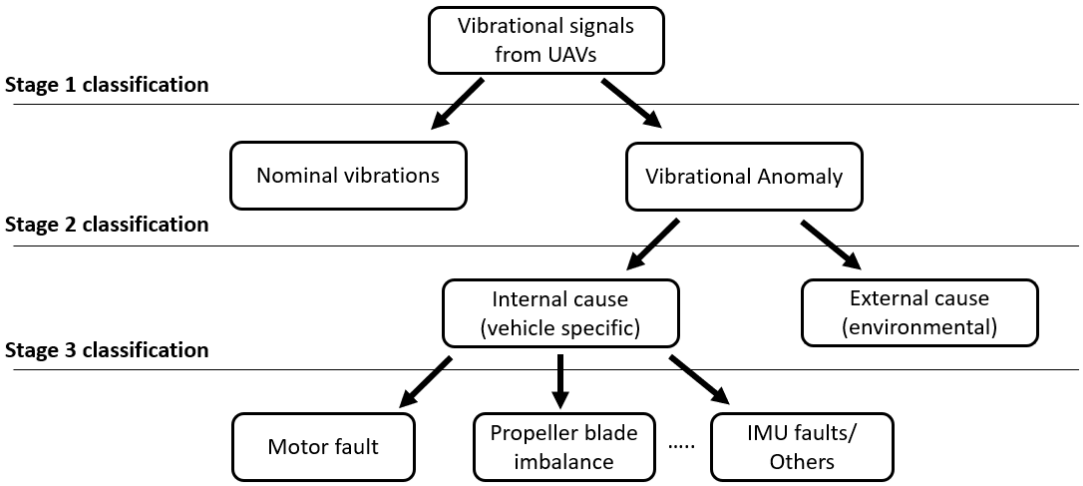


Figure 1: Classifying Vibration Signals

In this document, we explore the applications of frequency analysis for studying and classifying vibration signals due to wind. We begin with an introduction to the data, followed by background on frequency analysis techniques commonly used for studying and classifying mechanical vibrations, specifically those based on the Fourier transform, discussing their mathematical foundations and common issues that arise with them. We apply these techniques to vibrational signals obtained during four different test flights performed at Langley Research Center. We conclude with an overview of the challenges faced with this particular data set, as well as some potential next steps.

2 Data

In this section, we summarize the data used in this study; for further information on the data set, see [1, 4]. The data for this study was collected during four flight tests performed at Langley Research Center using an octocopter similar to *DJI S1000* specifications in autonomous mode and equipped with Pixhawk hardware and controlled with Ardupilot software. Figure 2 shows the aircraft used. Data collected by the aircraft include air-

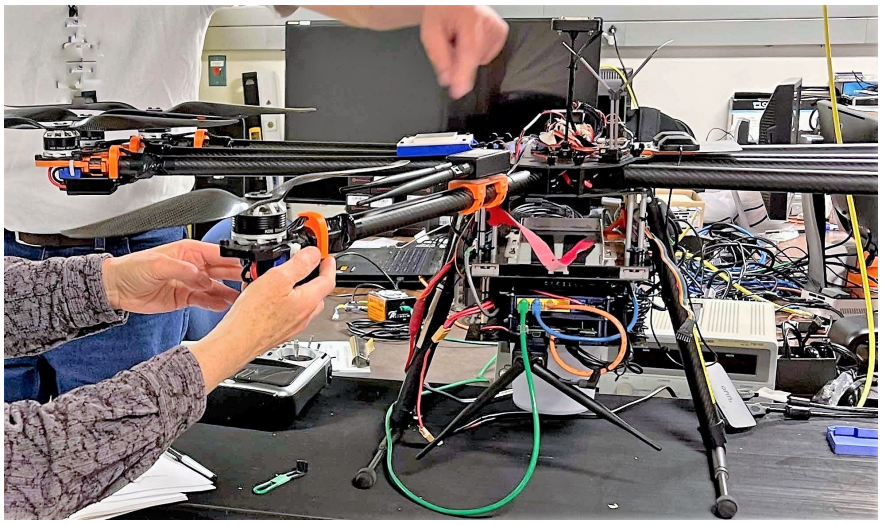


Figure 2: Octocopter used in Langley flight tests

craft status, such as power draw, position and heading information, IMU data from an accelerometer and gyroscope, and vibration data. All data obtained from the aircraft was sampled at a rate of $2Hz$. We note that these sensors are all contained in the center of the aircraft, and sensor data for the individual motors is not available. All four test flights followed the same trajectory with 9 waypoints, shown in Figure 3.

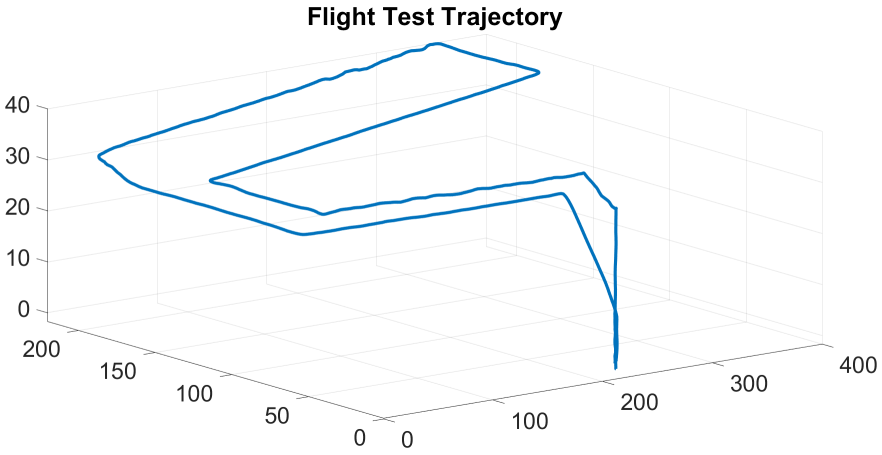


Figure 3: Trajectory

The flights had the following ground and wind speed profiles:

- **Flt 41:** low ground speed ($5m/s$) with mostly low wind speeds ($<2.5m/s$)
- **Flt 43:** low ground speed ($5m/s$) with mostly high wind speeds ($<6m/s$)
- **Flt 44:** high ground speed ($8m/s$) with mostly high wind speeds ($<5m/s$)
- **Flt 45:** high and low ground speeds ($8m/s$ and $3m/s$ during final segment of flight)

with mostly high wind speeds (<4.5m/s)

The wind speed data was collected by a Centralized PI Controller located within 100m of the flight path, and was sampled once every 5s. This data also included the heading of the wind, and we note that no information for a vertical component is included. The effects of surrounding trees and building is considered to be negligible. The wind speeds for each flight are shown in Figure 4.

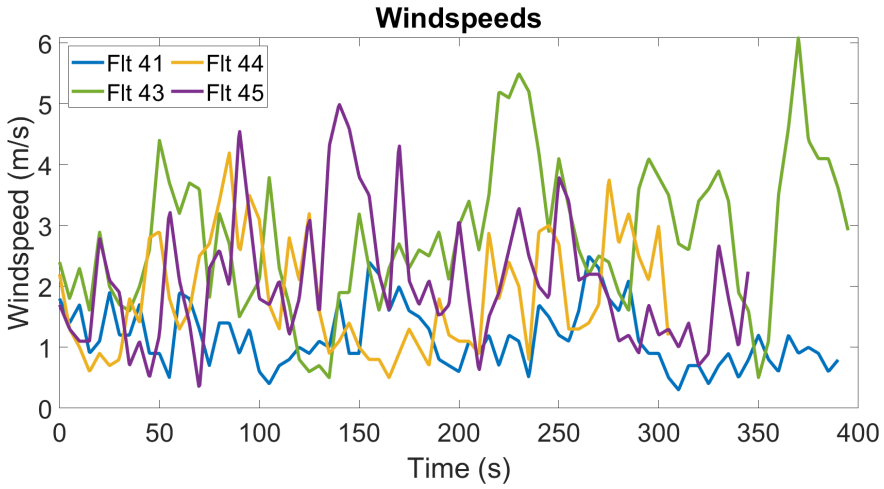


Figure 4: Wind speeds for all flights

The vibration data was collected for the x, y, and z axes of the aircraft, where the x and y axes are parallel to the body of the aircraft and the x axis points toward the front of the aircraft; the z axis then represents the vertical components. As previously mentioned, this data was collected at a rate of 2Hz. Collection began a few minutes prior to flight and concluded about a minute after flight. Since the data contained so many extra samples outside of flight, it had to be trimmed to only include the time window of interest. This time frame was determined based on when the z axis acceleration showed significant increase, indicating take off, and concluded when these values stopped changing, indicating landing. The vibration data for each axis from the core time frame is shown in Figure 5.

However, for most of the analyses, it was determined that the magnitudes of the vibrations for each flight would be preferable. This was due to the vibration signals in each axis having similar features. Moreover, the wind direction information is relative to a cardinal axis, which is a constant frame of reference, while the vibration axis information is based on the heading of the aircraft, which is variable relative to the axes used to measure wind direction. Thus, for the initial analysis, direction information was disregarded and a focus was placed on comparing wind and vibration magnitudes. The vibration magnitudes are shown in Figure 6.

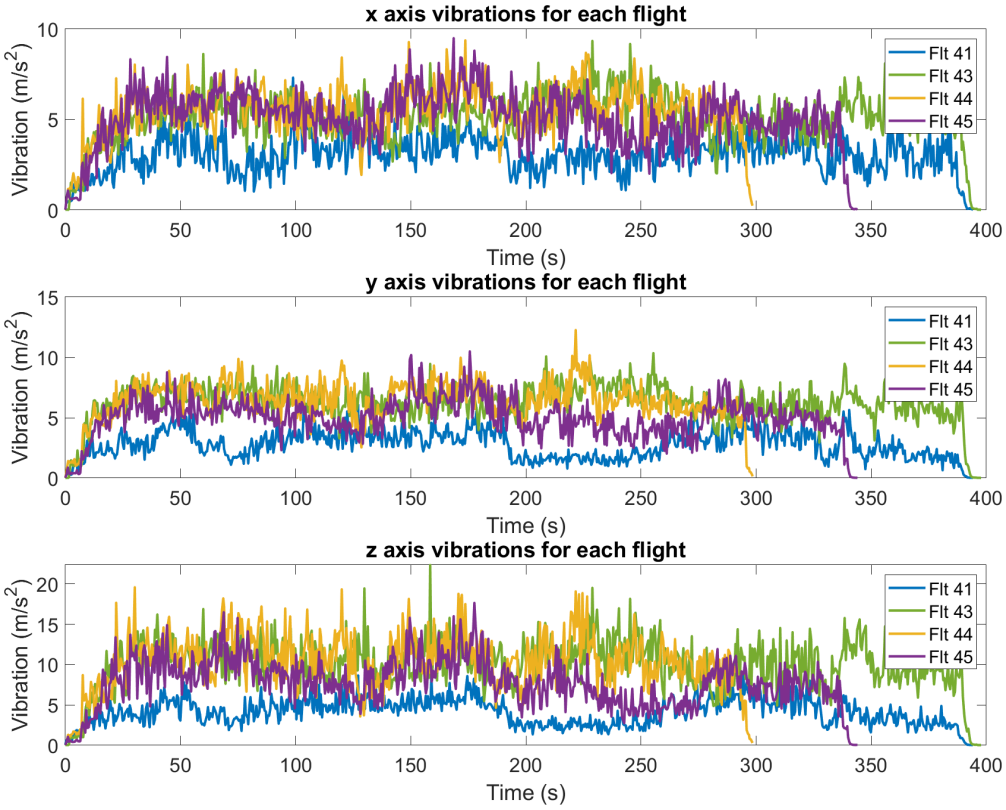


Figure 5: x, y, and z axis vibration data for all flights

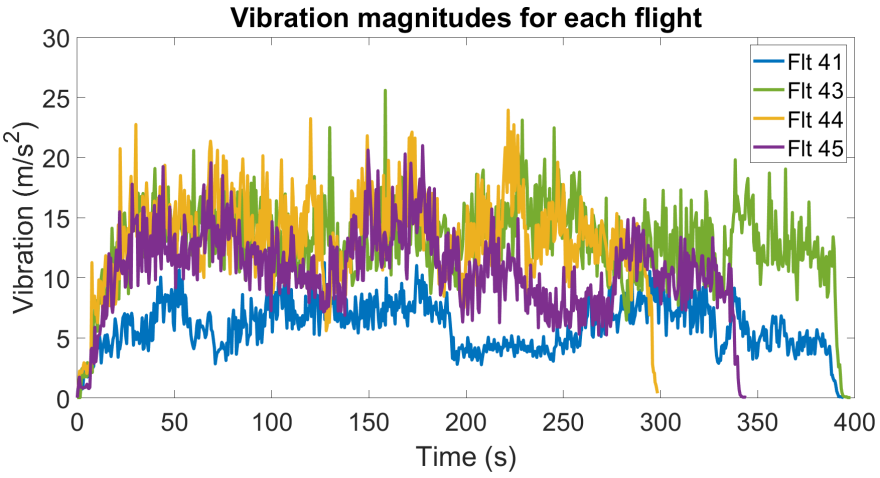


Figure 6: Vibration magnitude data for all flights

3 Methods: Mathematical background

In this section, we discuss the frequency analysis methods applied as well as their mathematical backgrounds. We begin with the foundation of each of these tools – the Fourier series. We also discuss challenges that arise when working with a discrete domain, such as aliasing and leakage, followed by a description of the Fast Fourier Transform (FFT), the algorithm used by MATLAB to compute the discrete Fourier transform. We then turn our attention to each of the methods used that are based on the discrete fourier transform, the power spectral density, Welch's method, windowing approaches, and the short time Fourier transform (STFT).

3.1 Fourier Analysis and the Discrete Fourier Tranform

We start with an overview of the derivation of the Fourier series and the discrete Fourier transform. For further details in the computations, please see [3, 14, 18]. The foundation of Fourier analysis lies in expressing a continuous, periodic signal, x, as an infinite linear combination of the form

$$x(t) = \sum_{k=0}^{\infty} a_k \cos(kt) + b_k \sin(kt). \tag{1}$$

While much of the work related to this area of mathematics is attributed to its namesake, Joseph Fourier during the 19th century, contributions were made to its advancement by Leonard Euler and Daniel Bernoulli as early as the 18th century [9]. Originally studied due to its natural applications in partial differential equations, it quickly became a popular tool in signal processing due to its ability to break complicated signals down into simple building blocks [18].

Note that on the right hand side of (1), the series is a periodic function of period 2π, so x must also have period 2π. Such a representation can be determined for any periodic function, but for ease of notation, we assume here the signal has period 2π. The choice of trigonometric functions to construct this series is partially based on their orthogonality, that is, the relationships,

$$\begin{aligned}
\int_{-\pi}^{\pi} \cos(kt)\sin(jt)dt &= 0, \quad \forall k, j \\
\frac{1}{\pi}\int_{-\pi}^{\pi} \cos(kt)\cos(jt)dt &= \begin{cases} 1 & k = j; \\ 0 & k \neq j; \end{cases} \\
\frac{1}{\pi}\int_{-\pi}^{\pi} \sin(kt)\sin(jt)dt &= \begin{cases} 1 & k = j; \\ 0 & k \neq j; \end{cases}
\end{aligned} \quad .$$

By multiplying the trigonometric series in (1) by $\cos(kt)$ and $\sin(kt)$ and applying elementary trigonometric identities, we deduce that the coefficients of the such a trigonometric series must be given by

$$a_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} x(t)dt$$
$$a_k = \frac{1}{\pi}\int_{-\pi}^{\pi} x(t)\cos(kt)dt$$
$$b_k = \frac{1}{\pi}\int_{-\pi}^{\pi} x(t)\sin(kt)dt.$$

The Fourier transform refers to this process of obtaining the coefficients corresponding to the frequency components of a signal. For example, consider the following signal x shown in Figure 7. This periodic signal can be written as

$$x(t) = 7 + 3\sin(\frac{1}{2}t) + sin(t) + 5\sin(3t) + 2\sin(11t),$$

and its components are shown in Figure 8.

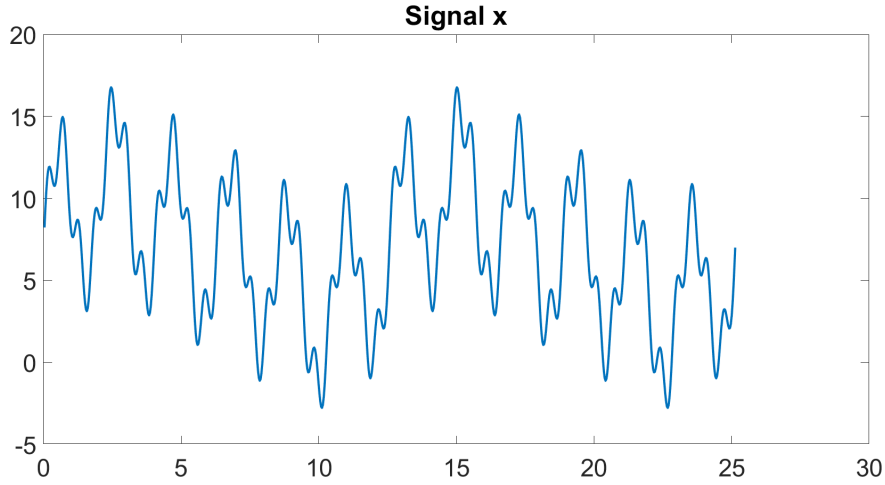


Figure 7: Example signal, $x(t) = 7 + 3\sin(\frac{1}{2}t) + sin(t) + 5\sin(3t) + 2\sin(11t)$

The Fourier transform tells us the magnitudes of the dominant frequency components present in the signal. which we call the frequency spectrum. This is shown in Figure 9. This agrees with what we know the signal to be composed of, with a constant term of magnitude 7 corresponding to frequency $k = 0$, the component of frequency $k = \frac{1}{2}$ has magnitude 3, the component of frequency $k = 1$ has magnitude 1, etc..

When working with the magnitudes of the frequency components, we often say that we are analyzing the signal in the frequency domain, while the original time series is

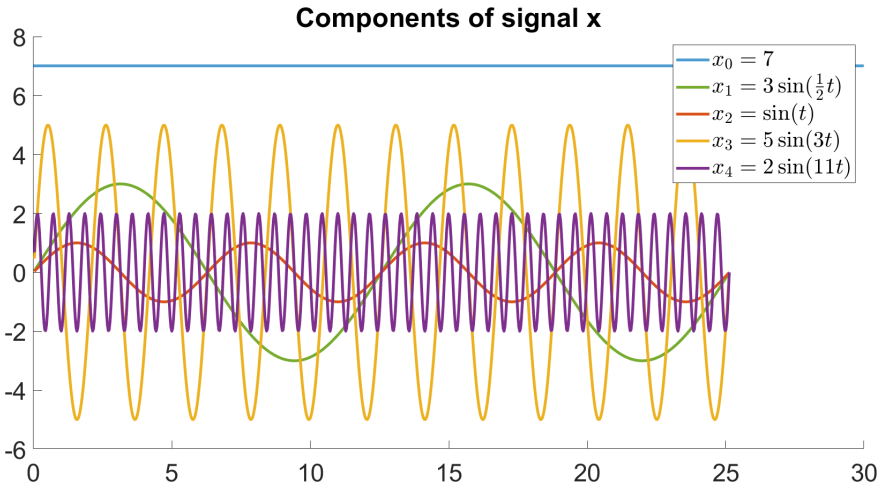


Figure 8: Components of x

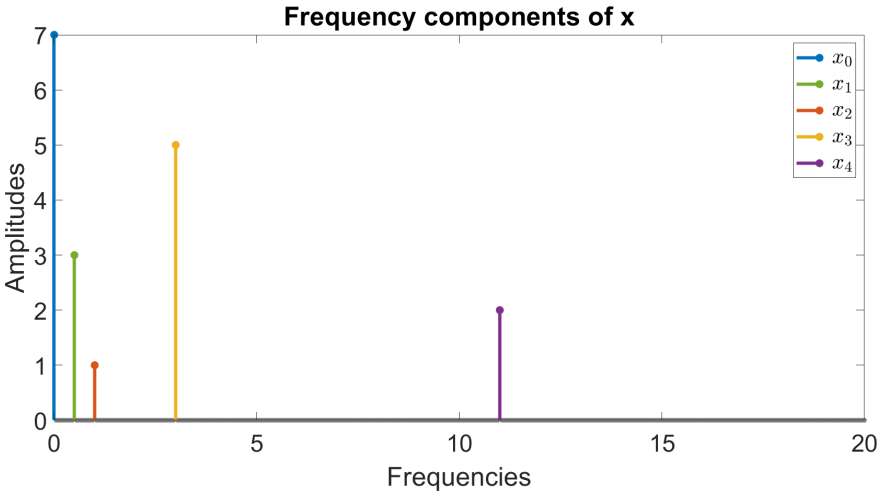


Figure 9: Frequency spectrum for x

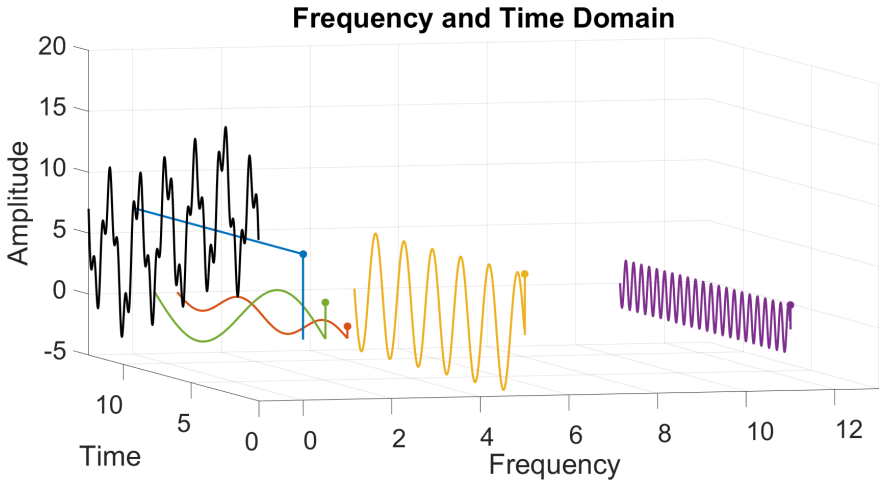


Figure 10: Components of x in time and frequency domains

considered to be in the time domain. Figure 10 shows the relationship between these two domains.

While breaking down a signal into sines and cosines offers significant advantages, an immediate inconvenience of this expression is that for each frequency k, there are two coefficients corresponding to it. Therefore, it is more common to merge the sine and cosine terms into a single exponential expression using Euler's formula:

$$e^{i\theta} = \cos(i\theta) + \sin(i\theta).$$

This relationship yields the following alternative expressions for sine and cosine:

$$\cos\theta = \frac{e^{i\theta} + e^{-i\theta}}{2}$$
$$\sin\theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}.$$

Substituting these expressions into (1), we obtain

$$x(t) = \sum_{-\infty}^{\infty} X(k) e^{ikt},$$

where

$$X(k) = \frac{1}{2\pi} \int_{-\pi}^{\pi} x(t) e^{-ikt}.$$

For convenience, it is common to rescale so the signal has period of 1, giving the following

expressions:

$$x(t) = \sum_{-\infty}^{\infty} X(k) e^{2\pi i k t},$$

with

$$X(k) = \int_0^1 x(t) e^{-2\pi i k t}, \tag{2}$$

where the integral can be shifted from bounds of $-1/2$ to $1/2$ to bounds of 0 to 1 due to symmetries arising from the periodicity of the signal. The amplitudes of the frequency components are then given by $|X(k)|$, where $|a + bi|$ represents the modulus of a complex number $a + bi$ given by $\sqrt{a^2 + b^2}$.

As previously mentioned, this mapping from a signal x to its corresponding Fourier coefficients is often referred to as the Fourier transform of x. This terminology also often refers to the mapping to obtain such coefficients for nonperiodic signals, which is also based on exponential expressions of similar structure; we refer the reader to [9, 18] for details on this case.

Note that taking the integral in (2) assumes the signal x to be continuous. However, in practice, signals cannot be sampled continuously and are considered discrete in time. With this in mind, we compute the coefficients $X(k)$ using a sum instead of an integral.

$$X(k) = \sum_{n=0}^{N-1} x(n) e^{\frac{-2\pi i k n}{N}}$$

where N is the number of samples, n is the sample number, and $x(n)$ is the signal at the nth sample. In the discrete case, the amplitudes are given by the normalized modulus of the coefficients, that is, $\frac{1}{N}|X(k)|$.

To reconstruct the sampled signal, we use the following expression:

$$x(n) = \frac{1}{N} \sum_{k=0}^{N-1} X(k) e^{\frac{2\pi i k n}{N}}. \tag{3}$$

Two important questions arise when working with a discrete signal 1) What rate should the signal be sampled at? 2) How many samples should be taken?

The answer to the first question lies in the sampling theorem. The sampling theorem tells us that the highest frequency in a signal that can be accurately represented through the discrete Fourier transform is half of the sampling frequency. That is, letting f_s represent the sampling frequency, the highest frequency in the signal that is accurately detected is $f_s/2$; any frequency present in the signal larger than $f_s/2$ will likely suffer aliasing, i.e.,

be misrepresented as a component of lower frequency. This cutoff of $f_s/2$ is called the Nyquist frequency. The issue with aliasing is illustrated in Figure 11.

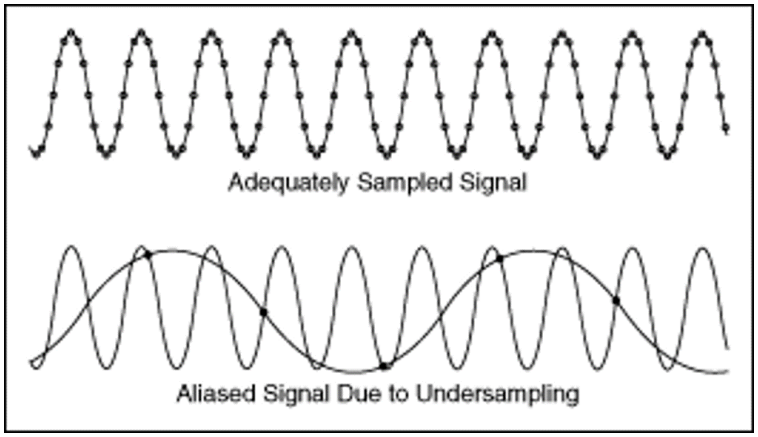


Figure 11: Slow sampling can result in aliasing (Image from National Instruments *Signal Processing and Analysis* [15])

Going back to our example signal, $x(t) = 7 + 3\sin(\frac{1}{2}t) + sin(t) + 5\sin(3t) + 2\sin(11t)$, the highest frequency present is $k = 11$. This means we must sample at a rate of at least $f_s = 22$ in order to accurately capture this frequency. If we sample at a lower rate, say $f_s = 20$, this frequency is still detected, but represented as a lower frequency, as shown in Figure 12. Indeed, the first four components are accurately represented, while the component corresponding to 11 is represented as 9.

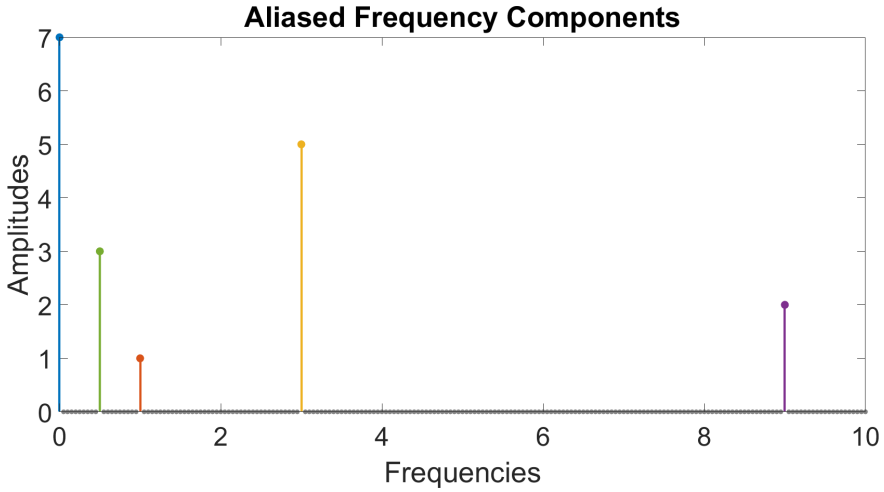


Figure 12: Sampling $x(t)$ at a rate of $f_s = 20$, leads the $k = 11$ component to alias as $k = 9$.

Thus, in order to obtain an accurate frequency spectrum, the signal must be sampled at a rate of $f_s \geq 2f_{max}$, where f_{max} is the highest frequency present in the signal.

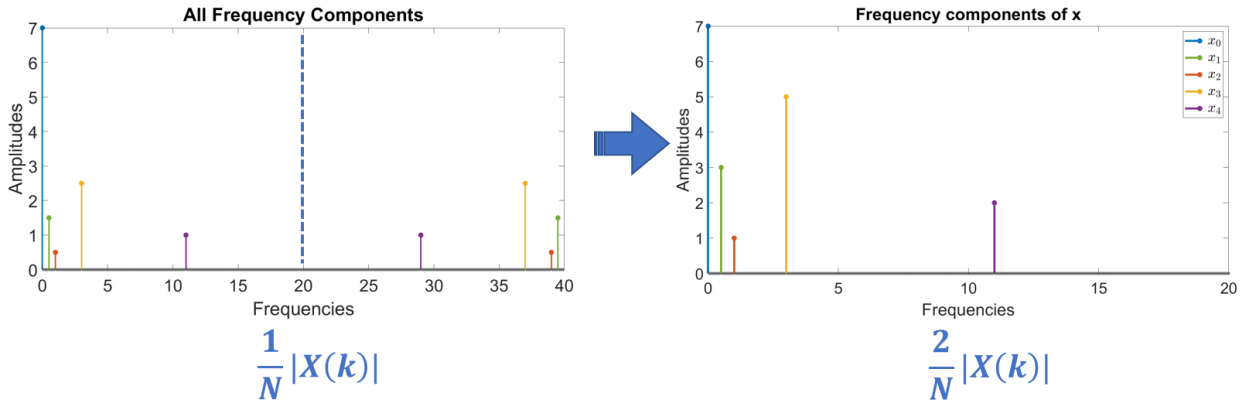


Figure 13: Using symmetry in frequency components to compute amplitudes

The answer to the second question, how many samples to take, is related to the frequency resolution, that is, the spacing between frequencies detected by the DFT. Note a consequence of (3) is that the number of frequencies detected by the DFT is equal to the number of samples used, N. As a result, the frequency resolution is given by $f_s/N = 1/T$, where T is the time duration of the sampled signal. Thus, once the sampling frequency is set, in order to achieve a desired frequency resolution, the number of samples, and therefore the duration of the signal, must be adjusted accordingly.

Frequencies detected beyond the Nyqist frequency represent negative frequency values, as there is a periodicity in the frequency components detected by the DFT [19]. In the case that the signal x is real, the corresponding coefficients are complex conjugates of each other. That is, if $X(k) = a + bi$, then $X(-k) = a - bi$. As a result, $|X(k)| = |X(-k)|$. In this instance, convention is to work with a one-sided DFT, where only positive frequencies are considered and their amplitudes are doubled. This is illustrated in the context of our earlier example in Figure 13.

A final potential issue to be aware of when working with the discrete Fourier transform is that of spectral leakage. Spectral leakage occurs when the signal is not periodic in the time frame it is sampled in. For example, our signal x is periodic in the sample shown, but if a shorter segment is taken, it introduces frequencies that would not otherwise be present due to potential discontinuities that arise and shortening the periods of the features, as shown in Figure 14. The result is that the frequency components "leak" into nearby components and their amplitudes may no longer be accurately represented, as shown in Figure 15

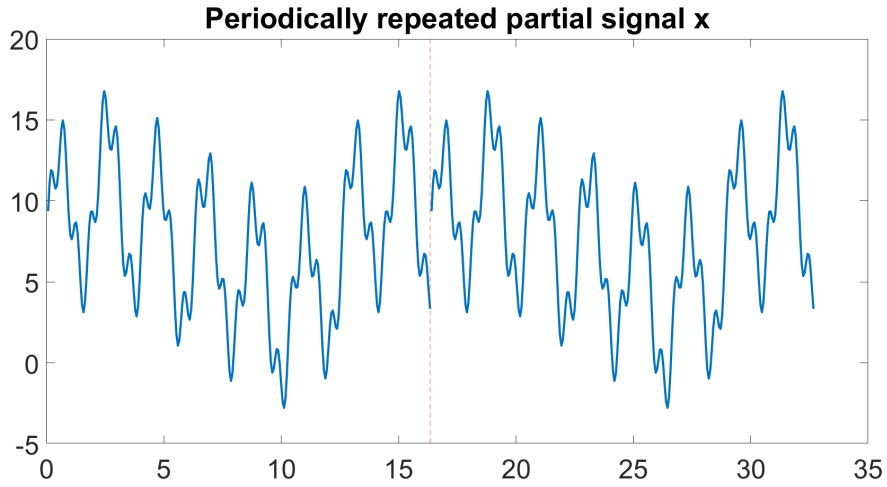


Figure 14: x periodically repeated in shorter time frame

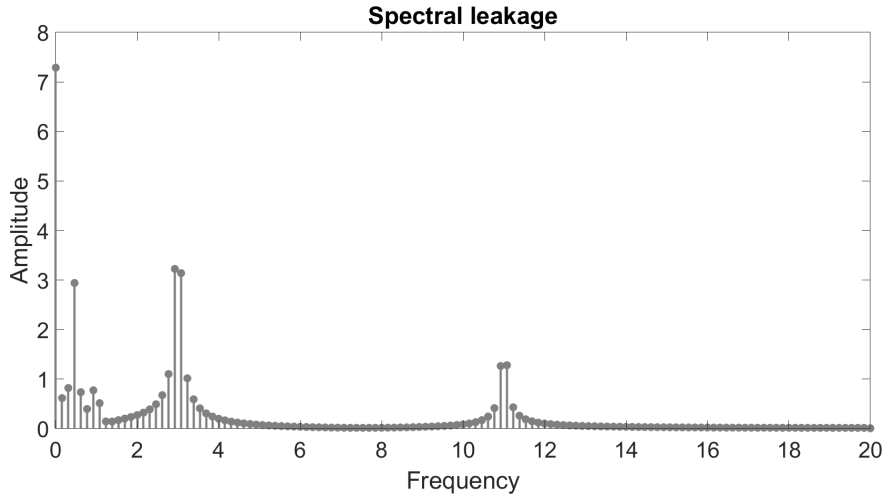


Figure 15: Spectral leakage resulting from sampling a time segment x is nonperiodic in

3.2 Fast Fourier Transform in MATLAB

In this section, we discuss the algorithm used the compute the DFT, the Fast Fourier Transform (FFT), and its application in MATLAB.

Computing the discrete Fourier transform is a straightforward process, as it is a sum of samples multiplied by an exponential term. It then has to perform N multiplications and N additions, resulting in $O(N^2)$ operations total. For larger problems, this can result in exceptionally slow computation times. To solve this, the Fast Fourier Transform was proposed by Cooley and Tukey in 1965 [8]. The main idea behind this algorithm is to exploit symmetries present in the exponential terms of the Fourier transform.

First, the sum is split between even and odd samples. That is,

$$\begin{aligned} X(k) &= \sum_{n=0}^{N-1} x(n) e^{\frac{-2\pi i n}{N}} \\ &= \sum_{n=0}^{N/2-1} x(2n) e^{\frac{-2\pi i n}{N/2}} + e^{\frac{-2\pi i n}{N}} \sum_{n=0}^{N/2-1} x(2n+1) e^{\frac{-2\pi i n}{N/2}} \\ &= X_{even}(k) + X_{odd}(k). \end{aligned}$$

An immediate consequence of this is that the term $e^{\frac{-2\pi i n}{N/2}}$ now only needs to be computed half as many times, and these two sums can be computed in parallel. We also recall that

$$e^{-2\pi i n} = 1,$$

and therefore

$$e^{\frac{-2\pi i (k+N/2) n}{N/2}} = e^{\frac{-2\pi i k n}{N/2}},$$

meaning that

$$X_{even}(k + N/2) = X_{even}(k) \quad \text{and} \quad X_{odd}(k + N/2) = X_{odd}(k).$$

Thus X_{even} and X_{odd} only need to be computed for $k < N/2$. From here, the computed terms can be mixed and matched to produce the coefficients corresponding to each frequency. The "butterfly" diagram in Figure 16 (image from Wikipedia, the free encyclopedia [21]) demonstrates this for an example computing 8 terms.

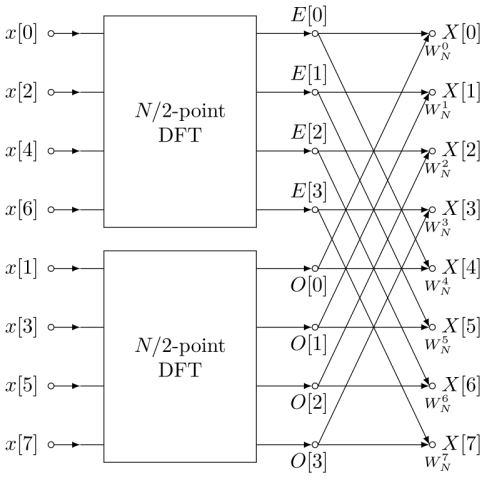


Figure 16: FFT butterfly diagram

However, we do not have to stop at splitting the sum once; we can continue to split these

sums until only a couple of specific terms need to be computed. The resulting number of computations is now $O(N \log_2 N)$.

MATLAB computes the DFT using the FFT. Below is an example of the code used to compute the DFT in MATLAB for Flight 41.

```
% v41m represents the vibration magnitudes from flight 41
Fs41 = 2;   % Sampling frequency
T41 = 1/Fs41;   % Sampling period
L41 = length(v41m);   % Length of signal

% Compute DFT; this returns the complex valued coefficients X(k)
ftv41m = fft(v41m);

% take |X(k)|; normalize by number of samples, L41
ftv41m_amp = abs(ftv41m/L41);

% select frequencies below Nyquist frequency;
% use (L41+1)/2 when sample length is odd
ftv41m_amp1 = ftv41m_amp(1:L41/2 + 1);

% double coefficient magnitudes for repeated frequencies;
% first and last frequencies are not repeated
ftv41m_amp1(2:end-1) = 2*ftv41m_amp1(2:end-1);
```

To display the magnitudes of the frequency components, use the following:

```
% frequency components for x axis
f_window41 = Fs41*(0:L41/2)/L41;

% plot frequency amplitudes
figure;
plot(f_window41, ftv41m_amp1)
title("Flt 41: FFT")
xlabel("Frequency")
ylabel("Amplitude")
```

3.3 Power Spectral Density and Welch's Method

In this section, we discuss the power spectral density (PSD) and an averaging approach based on it called Welch's method.

The power spectral density indicates the power contained in each frequency component. It is computed from the coefficients of the DFT and normalized to indicate the power each frequency contributes to the signal [6, 13], and is given by

$$P(k) = \frac{1}{N}|X(k)|^2,$$

where $X(k)$ is as given in (2), and has neither been normalized nor doubled to account for symmetries yet. The peaks of the PSD of a signal occur at the same frequencies as the traditional DFT, but are scaled differently which can dramatize their appearance. To compute the PSD in MATLAB, the following code can be used:

```
% v41m represents the vibration magnitudes from flight 41
Fs41 = 2;   % Sampling frequency
T41 = 1/Fs41;   % Sampling period
L41 = length(v41m);   % Length of signal

% compute PSD using the FFT of v41m indicated above;
% square and normalize the amplitudes from the DFT
psd_v41m_amp = abs(ftv41m).^2/L41;

% select frequencies below Nyquist frequency;
% use (L41+1)/2 when sample length is odd
psd_v41m_amp1 = psd_v41m_amp(1:L41/2+1);

% double values for repeated frequencies;
% first and last frequencies are not repeated
psd_v41m_amp1(2:end-1) = 2*psd_v41m_amp1(2:end-1);
```

To plot the PSD, use the same code as from the DFT.

Sometimes, the resulting data in the frequency domain is noisy. When this occurs, it can be helpful to smooth out the results using an averaging approach called Welch's method. Welch's method divides the data into smaller, overlapping segments (Figure 17), applies a windowing function to each (Figure 18), takes the PSD of each segment, then averages them together.

In particular, if M windows of N samples each are to be used, for the mth window, compute the PSD incorporating the windowing function:

$$P_m(k) = \frac{1}{N}|X_m(k)|^2,$$

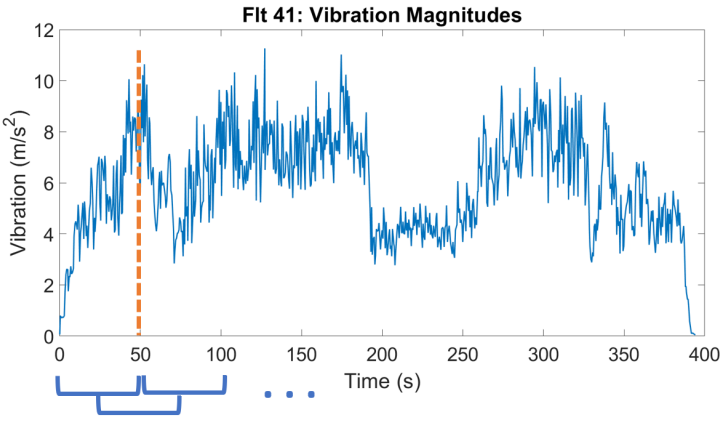


Figure 17: Divide data into overlapping segments

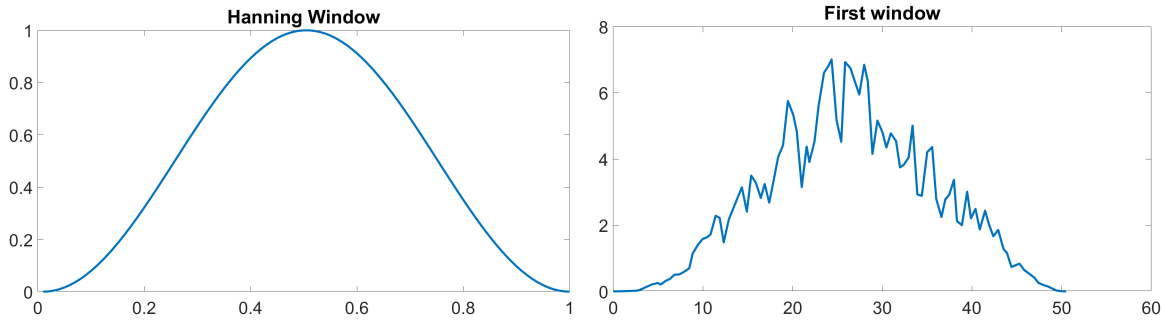


Figure 18: Apply windowing function to each segment of data through multiplication

where $X_m(k)$ is the DFT incorporating the windowing function in its computation, namely

$$X_m(k) = \sum_{n=0}^{N-1} x_m(n)w(n)e^{\frac{-2\pi ikn}{N}}$$

($w(n)$ is the windowing function evaluated at the nth point of time in the window under consideration). Welch's periodogram is then given by the average of the PSD's across all segments of data, that is,

$$S(k) = \frac{1}{M}\sum_{m=0}^{M-1} P_m(k).$$

Implementing this process in MATLAB is straightforward; MATLAB includes a command *pwelch* which automatically performs all of these computations.

3.4 Windowing Approach and the Short Time Fourier Transform

A major drawback of working in the frequency domain is the loss of time information. To mitigate this, it is common to take the DFT of smaller regions of the data.

The first approach that will be used is to simply divide each data set into smaller segments, take the DFT of each segment, then take note of the largest peaks present in the frequency spectrum for each window. This allows the frequency information to be connected to specific time frames within the data set. However, while successful in certain applications [5], it does not offer high resolution in time.

A popular approach to this is the short time Fourier transform (STFT), which is similar to the windowing approach described above in that it takes the DFT of windows of the data. It achieves higher time resolution by allowing the windows to heavily overlap and applying a windowing function similar to that shown in Figure 18 to emphasize the data contained in the center of each window and de-emphasize the data contained toward the edges of the window, which would overlap with neighboring windows. Similar to Welch's method, the DFT in each window is given by

$$X_m(k) = \sum_{n=0}^{N-1} x_m(n) w(n) e^{\frac{-2\pi i k n}{N}}.$$

Once the DFT is computed for each window, the time and frequency information is plotted together, with a color scale indicating the magnitudes of the frequencies components present in each window of data. Computing the STFT in MATLAB is also simple, as it has the command *stft* available.

A disadvantage to the STFT is the tradeoff between the frequency resolution and time resolution. As stated previously, the frequency resolution is given by $f_s/N = 1/T$, where T is the duration of time the sample is taken for. This means that there is an inverse relationship between the frequency resolution and the time resolution available with the STFT; that is, decreasing the window size to improve time resolution necessarily worsens frequency resolution, and to increase frequency resolution, the window size must be increased, worsening time resolution [10].

4 Results

In this section, we discuss the results obtained through the methods discussed above. We begin with some initial observations in the time domain, then discuss the results obtained using frequency domain techniques that disregard time information, as well as time frequency domain techniques that are time dependent.

4.1 Time Domain Analysis: Initial Observations

In the time domain, we first consider the summary statistics for the wind and vibration data. For this, we focus on Flight 41 and Flight 43, as they both had the same ground speed at 5m/s, but the wind speed during Flight 43 was much higher than Flight 41.

Figure 19 shows plots and histograms of the vibration and wind speed magnitudes for only Flight 41 and Flight 43. It can be seen that the vibration magnitudes in Flight 43 are much higher than Flight 41. In particular, in Figure 19d, it is shown that the average wind speed during Flight 43 was 2.82m/s, more than twice that during Flight 41, at 1.15m/s. In Figure 19c, the average vibration magnitude from Flight 43, at 12.86 m/s^2, is over twice that of Flight 41, at 5.95m/s^2. While a strict direct relationship does not follow from these observations (consider Flight 43, time 100 - 150 s, where the wind speed dips dramatically, but the vibration does not), it does suggest that there might be a correlation between the wind speed and resulting vibrations in the aircraft. With this in mind, we turn to the frequency domain in an effort to detect changes in any periodic behavior present in the vibration signals.

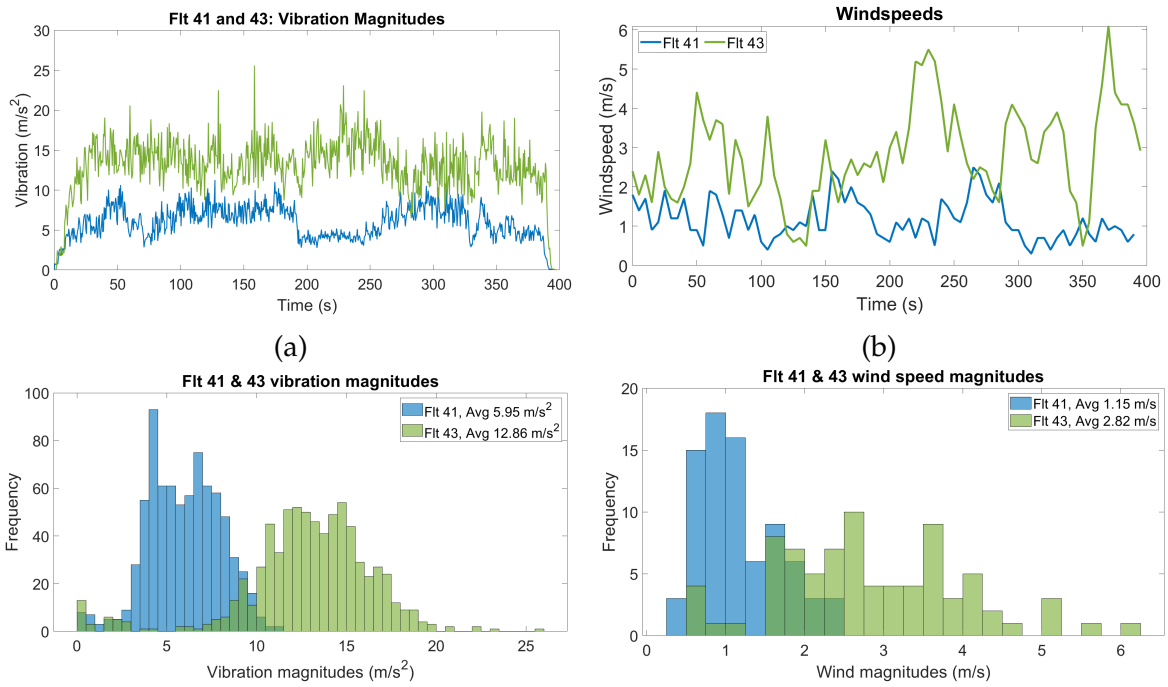


Figure 19: Vibration and windspeed magnitudes for Flight 41 and Flight 43

4.2 Frequency Domain Analysis: Time-Independent Methods

The first approach to working in the frequency domain is to simply take the DFT of the entire vibration signals under consideration to see if there are any immediately clear features present. We first consider results from the standard DFT, then look at results using the PSD and Welch's method.

We begin by taking a DFT of the vibration signals; Figure 20 shows the frequency spectrum for the vibration magnitude signals from each flight. The results shown are on a logarithmic scale, as the magnitudes of the frequency components were small compared to the magnitude of the initial frequency at zero. With the exception of the frequency zero, there are no notable peaks present in the frequency spectra from any of the flights. The spectra appear noisy without frequencies that stand out.

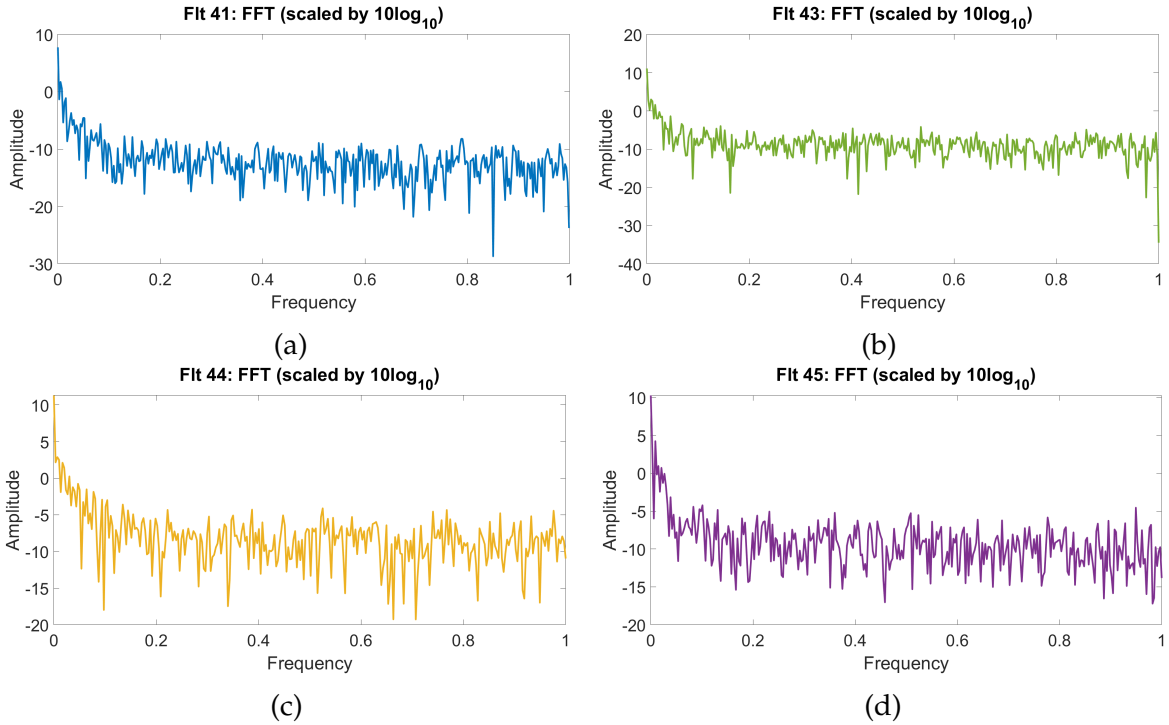


Figure 20: Frequency spectrum for all flights

Since the power spectral density offers different scaling from the DFT, this approach was also explored to extract features in the frequency domain. Figure 21 shows the PSD for each flight using a logarithmic scale. Similar to the results shown in Figure 20, the frequency spectra are noisy and exhibit no distinct features.

Finally, to smooth out the data in the frequency domain, we use the averaging approach offered by Welch's method. Figure 22 shows the Welch's periodogram for vibration data from each of the flights.

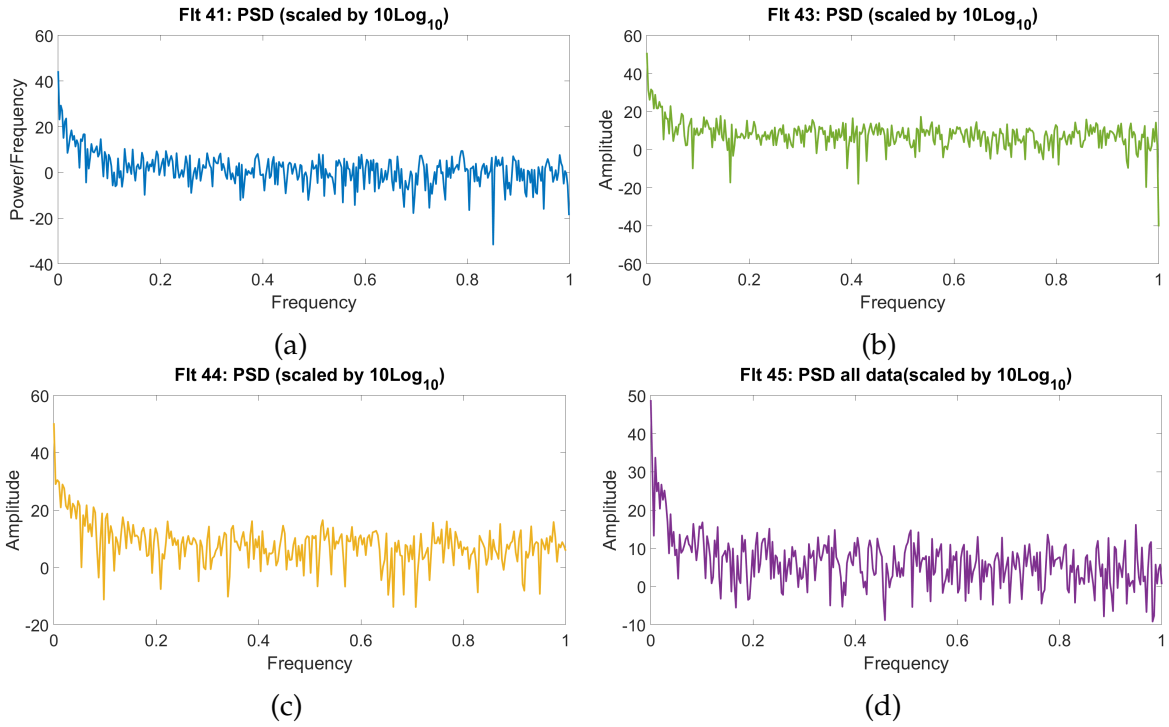


Figure 21: Power spectrum for all flights

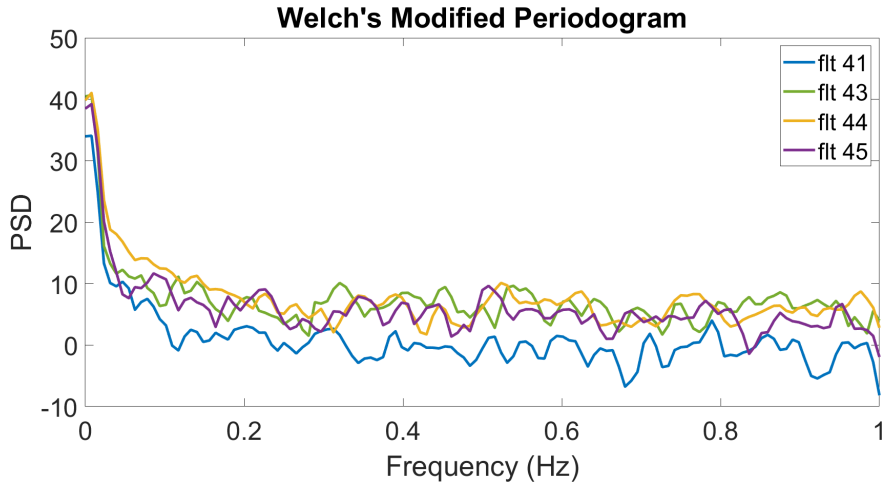


Figure 22: Welch's periodogram for all flights

Using this approach, we see that the spectra have smoothed out significantly and many of the peaks have consolidated. While there are no specific peaks that stand out, on this graph it is apparent that Flight 41, with the lowest wind speeds, has the lowest amplitude frequencies out of all the flights. While this does not provide information unique to the frequency domain, this does reinforce the observation in the time domain of a correlation between wind speed magnitude and vibration magnitude.

For each of these methods – DFT, PSD, and Welch's – no specific features appear in the frequency domain. However, it was noted that the amplitudes of the frequencies present in the vibrations from Flights 43, 44, and 45, were higher than those from Flight 41. This agrees with the previous observation that higher winds appear to correspond to higher overall vibrations during the flight.

4.3 Frequency Domain Analysis: Time-Dependent Methods

As mentioned in Section 3, a major drawback in applying tools based on the DFT is the loss of time information. To connect information in the frequency domain to segments of the flight of specific wind speeds, we apply two time dependent approaches to frequency analysis – windowing and the short time Fourier transform.

In the windowing approach, the vibration data from each flight was broken down into segments of 100 data points. Each flight was padded with zeros as needed to guarantee segments of length 100, so as the standardize the resulting frequency resolution when taking the DFT. For example, Figure 23 shows the segments used in Flight 41. Proceeding

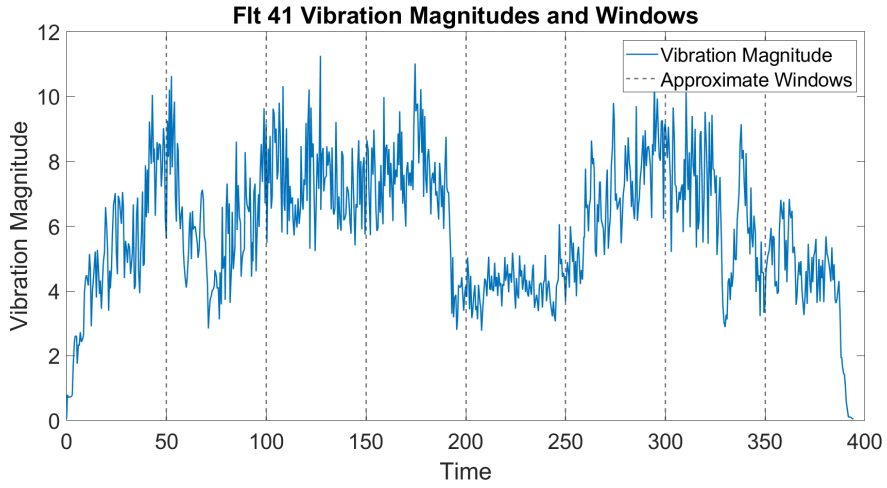


Figure 23: Flt 41 vibration data time segments

as in Section 3, we obtain the peak frequencies present in the spectra from each such window from each flight. These frequencies are plotted together showing frequencies and their corresponding amplitudes; the markers are color coded according to the wind speeds in their respective windows. Figure 24 shows these frequency data points color coded according to the maximum wind speed during each window of time, and Figure 25 shows them color coded according to the average wind speed.

In both figures, it can be seen that the peaks corresponding to windows with lower wind speeds tend to accumulate toward the bottom of the plot, while peaks corresponding with

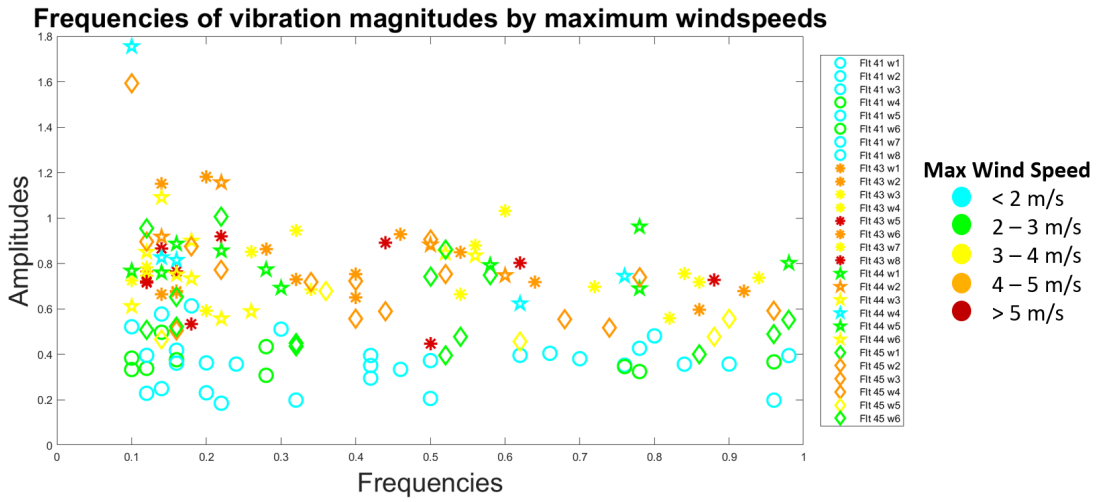


Figure 24: Peak frequencies for subsets of vibration data based on max wind speed

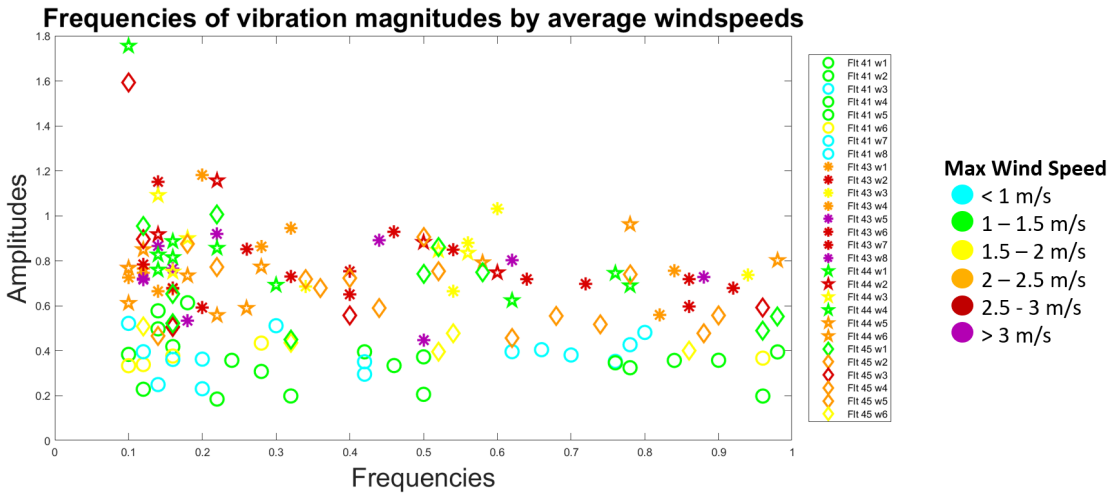


Figure 25: Peak frequencies for subsets of vibration data based on average wind speed

higher wind speeds tend to accumulate toward the top. This agrees with the observation made in Section 4.2 that higher wind speeds tend to correspond to increased amplitudes in the frequency domain. Ideally, if there had been any specific frequencies linked to higher wind speeds, we would also see peaks corresponding to higher wind speeds accumulating around those frequencies. However, both higher and lower wind speeds seem to correspond to relatively uniformly distributed frequencies.

Since the Welch's method heavily reduced the noise and number of peaks in the frequency domain, this analysis was repeated with taking the Welch's periodogram of each window, and using all the peaks present. Figure 26 shows the results from this approach, with the

plotted peaks color coded by maximum wind speed; a similar analysis was performed with the average wind speeds, but did not reveal any new features.

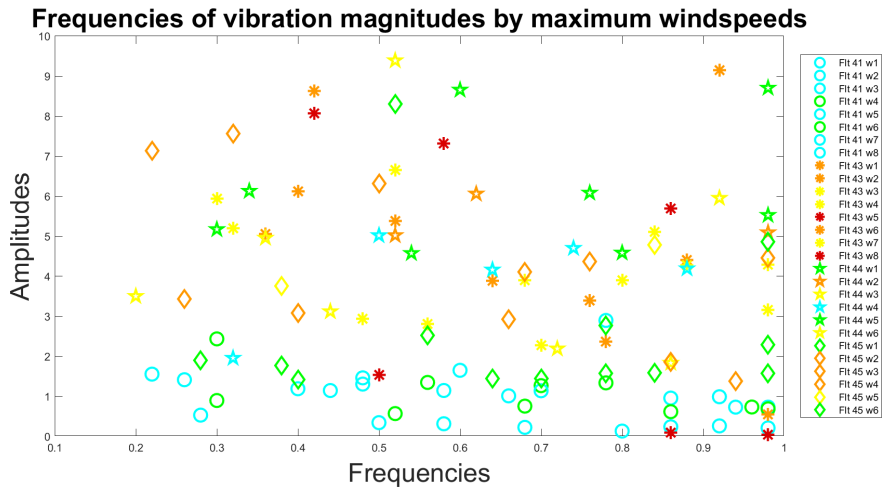


Figure 26: Peak frequencies for subsets of vibration data based on max wind speed

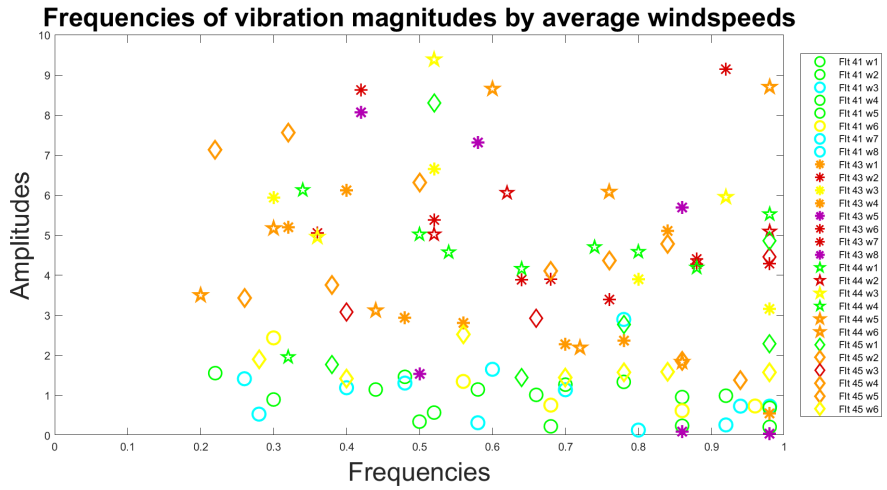


Figure 27: Peak frequencies for subsets of vibration data based on average wind speed

We see in these plots that, similar to Figure 24 and Figure 25, the peaks corresponding to the time periods with lower wind speeds tend toward the bottom, with the peaks corresponding to the higher wind speeds tending toward the top. However, these trends are not as distinct as in Figure 24 and Figure 25, with there being much more overlap between the points representing windows with high wind speeds and the points representing windows with low speeds. For example, in Figure 26, we see that the points representing some of the highest wind speeds are appearing at the very bottom of the plot. Once again, we also do not see any distinct clusters of points around specific frequencies.

Both analyses above, based on the traditional DFT as well as the Welch's modified PSD,

were repeated with the individual component information, i.e., vibration data along the x, y, and z axis. This approach did not yield any different features than those noted above.

Our final approach in the frequency domain was the short time Fourier transform. The short time Fourier transform was applied using a Hanning window with 64 samples and 75% overlap in the windows. Figure 28 shows the STFT for each flight using a logarithmic scale, with an overlay showing the maximum and average wind speeds in each of the windows of 100 data points used previously.

Once again, there appears to be some correlation between the amplitudes of the frequencies and the wind speeds. For example, in Flight 41, the amplitudes of the frequencies are higher in the regions where the wind speed is higher, with clear decrease specifically in the 200 - 250 s region where the wind speed drops. Similarly, Flight 43 shows higher magnitudes across the entire time frame, corresponding to the overall higher wind speeds. Finally, Flight 45 sees a drop in amplitudes after approximately 200 s, along with the drop in wind speed. However, this is also around the same time that the aircraft decreased its ground speed to 3 m/s, so it may also be worth considering the effects of changes in aircraft speed in this particular instance.

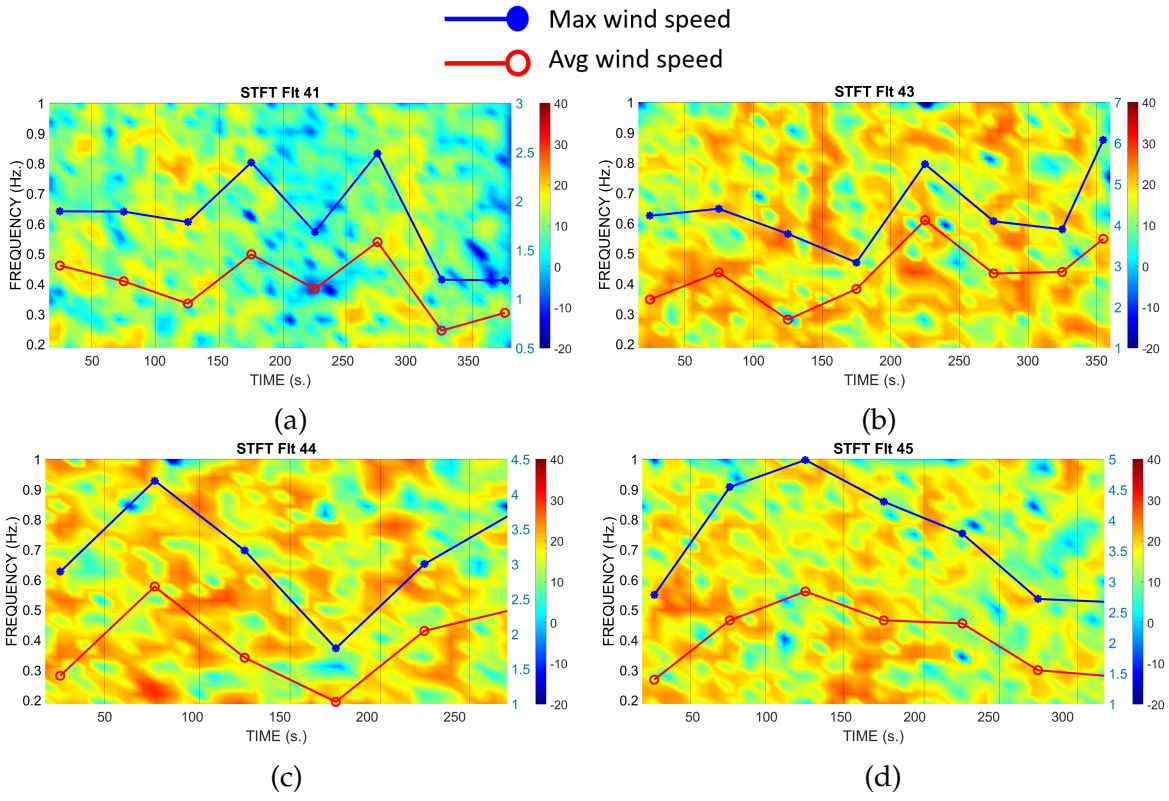


Figure 28: STFT with wind speed overlay for all flights

5 Conclusion

Throughout this analysis, we explored several different approaches to understanding UAV vibration signals in the frequency domain. We worked with the traditional discrete Fourier transform and power spectral density, as well as an averaging method, Welch's method, to smooth out the information contained in the frequency domain. Since the objective is connect changes in the frequency domain to changes in the wind speed, it was critical to also approach the analysis with methods that incorporated time dependence. Therefore, windowing approaches and the short time Fourier transform were also applied. In all of the methods used, no specific features were extracted in the frequency domain. However, the results did support the initial observation that overall increases in wind speed correspond to overall increases in vibration during flight.

In the remainder of this section, we discuss the challenges posed by the data set that may have led to the inconclusive results, as well as next steps in the analysis and future possible ways to collect data that could reveal more information.

5.1 Challenges

We now identify the challenges we faced with the data set that may have made it difficult to extract specific features in the frequency domain.

The largest issue faced was that of the sampling rate. The sampling rate came with two potential problems. The first is the sampling rate itself. A rate of 2Hz is very slow relative to the frequencies that should normally be present in the data; for example, the motors operate at a range of ???. Sampling at a frequency below double the frequencies expected to be present leads to aliasing, as indicated in Section 3. Moreover, the aircraft controls can react and adjust to changes in the wind in well under half of a second, indicating that valuable information on how the aircraft itself is responding to wind is lost in the gaps of time in between the data. Many other such studies are utilizing data sampled at a significantly higher rate to be able to capture this type of information [2, 7, 20].

In addition, the data was only sampled at 2Hz on average; this sampling rate was not constant throughout the data collection. With irregular sampling, it is possible that periodic behavior is being missed, which could case noisier data in the frequency domain. The slow sampling rate further compounds this issue, causing increased variability in time between samples.

Finally, the vibration data collected is noisy, as sensor is located at the center of eight motors that rotate independently. A vibration signal picked up from one of them would be mingled with the signals from the others, and since their RPM's are constantly, and independently, changing, this limits the distinct frequencies we can detect.

5.2 Next steps

The continued support of the trend between wind speed and vibration magnitudes is encouraging, and many other steps can be taken to better understand this connection.

An easy next step would be to perform the same analyses above using accelerometer and gyroscope data, which would offer data to work with that might have less noise from internal influences, such as the motors. However, such an analysis might not be particularly helpful as the measurements from the IMU's may not capture the periodic behavior of the motors, limiting the effectiveness of an analysis in the frequency domain. However, there is some potential for understanding vibration through accelerometer data analysis; [7, 20] both use accelerometer data for their vibration analyses.

Another option would be to perform the windowing analysis with windows based on types of segments of flight such as take off, landing, cruising, turns, etc.

Alternatively, techniques based on wavelet analysis could prove to be a better tool than those based on the Fourier transform. A wavelet transform uses wavelets as the basis functions instead of sines and cosines [10]. A wavelet $\psi(t)$ is a waveform that is nonzero only for a limited time and has an average value of zero; that is

$$\int_{-\infty}^{\infty} \psi(t)dt = 0.$$

For a signal x, wavelet decomposition of a signal is given by

$$D_x(a,k) = \sum_{n=0}^{N-1} x(n)\psi(a,k,n),$$

where a is a scaling term, k is a space/temporal term, and n is the sample. An example of a wavelet is shown in Figure 29, but wavelets can be a variety of shapes depending on the needs of the problem.

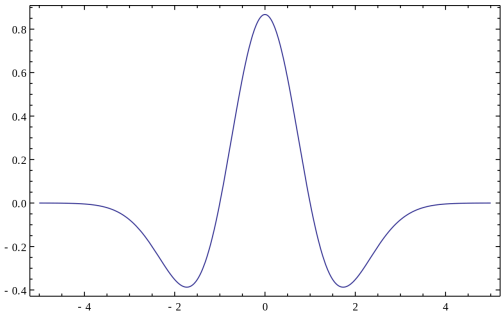


Figure 29: Wavelet example

A major advantage of the wavelet transform over the Fourier transform is its ability to provide *local* information about the signal [12]. Since wavelets are only nonzero locally, it can provide information about what is happening within the signal with greater time resolution than even the STFT can [2]. Indeed, wavelet analysis was introduced specifically to work with transient signals that have choppier, highly local characteristics [10, 12]. Moreover, a study done by [17] suggests it may also reduce any potential issues resulting from spectral leakage.

Several studies in machine fault detection have used wavelet analysis to detect anomalies in vibrations [2, 10, 17]. The nonperiodic, temporally-local nature of wind may make a wavelet approach more informative than the nonlocal Fourier transform. However, this approach could suffer from the same challenges with the current data set that the Fourier-based approaches did. If these issues are resolved in future data sets, a wavelet approach could be particularly helpful in detecting the relationship between wind and vibration signals.

MATLAB offers a wide variety of tools to perform wavelet analysis through their Wavelet Toolbox. We note that this toolbox is not included in the standard download of MATLAB and must be purchased separately.

Other directions for further work include considering the variance of the wind as well, to understand how changes in wind speed and direction may induce different vibrations in the aircraft. Some initial work in this direction was performed in the time domain, as well as using the windowing approach in Section 4.3. Figure 30 shows a comparison of the wind direction versus the aircraft heading by taking the cosine of the angle between these two headings, as well as the dot product wind velocity and the aircraft velocity. Figure 31 shows the vibration in each axis, as well as the magnitudes, with an overlay of the cosine of the angle between the aircraft and wind headings, θ_{diff}. The black dashed lines indicate a crosswind versus a headwind or tailwind, with the wind being a tailwind when $-45° < \theta_{diff} < 45°$, a crosswind when $45° < \theta_{diff} < 135°$ or $225° < \theta_{diff} < 315°$, and a headwind when $135° < \theta_{diff} < 225$.

Looking at the strong, lasting dip in vibrations during the time segment 200 - 250 s, it looks like there might be some correlation between the vibrations and if the wind is mostly a head or tail wind versus a crosswind.

When looking simply at the changes in wind speed, the same windowing analysis was performed, but it did not yield any different features than those in Figure 24 or Figure 25.

It would be particularly helpful to have more controlled wind information to use in these comparisons, so wind tunnel testing could be valuable. For example, measuring vibrations in the aircraft under steady winds at different speeds could provide baseline information for how the motors respond to headwinds of different strengths. The direction could also be varied for each test to understand better how vibration changes in a crosswind versus

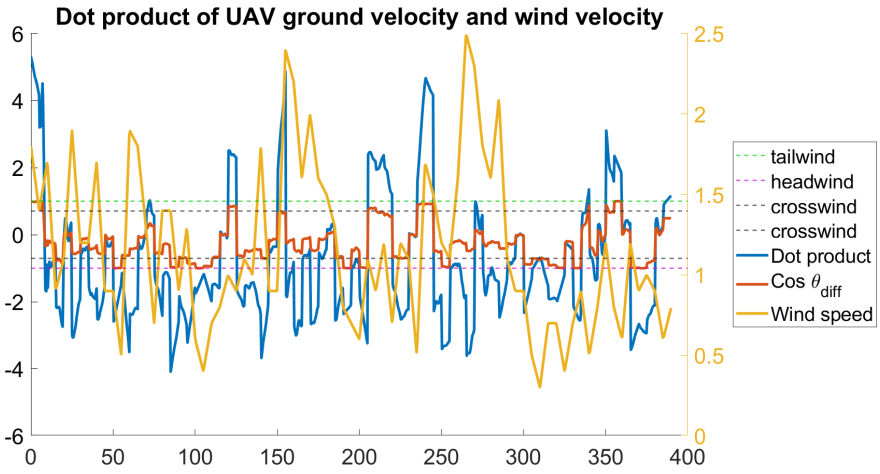


Figure 30: Dot product between wind and aircraft velocities

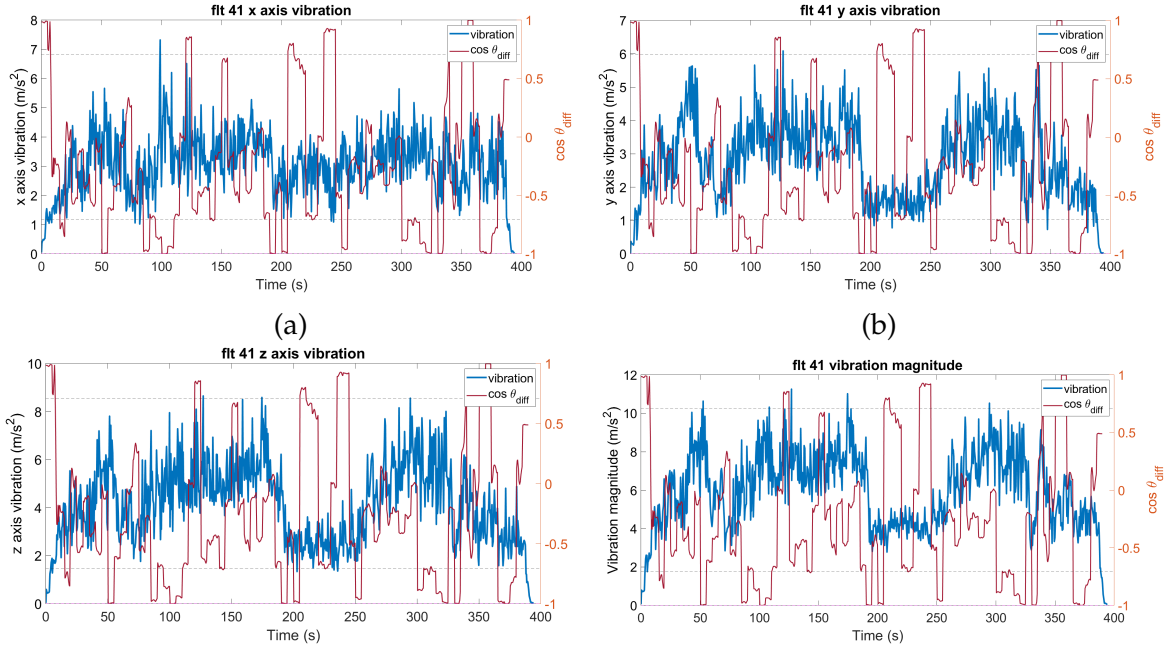


Figure 31: Vibration data with wind type overlay

a head or tail wind. Finally, it would be informative to have information about how the aircraft responds in changing winds, as changes in wind speed result in more turbulence than having a constant high wind.

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